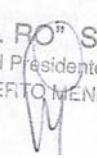
	CE COMPLIANCE DECLARATION DC1
---	--

I the Subscriber **Mennucci Lamberto**, in my quality of legal representative of the Factory ME.RO S.p.A. manufacturer of the machine as identified by the references below, declare that the said machine complies with the essential safety requirement and electromagnetic compatibility as stated in the European Community directives **2006/42/EC** and **89\336**, in particular the requirements of the following normatives have been accomplished : **UNI EN ISO 12100-1 , UNI EN ISO 12100-2 , UNI EN 13857 , UNI EN 349 , UNI EN 1088, CEI EN 60204-1 , EN 55011 , EN 50082-2.**

Manufacturer	Factory ME.RO S.p.A. Via Balestreri 430-Ponte a Moriano-LUCCA-ITALY-
Commission	5110\13 BOBST MANCHESTER
Machine	Electronic Generator
Type	MR/VP 20K IGBT
Serial number	n° 130000\C.5110\13
Year of construction	2013

It is forbidden to the end user to install the above stated machine within others without having verified the compliance with the directive 2006/42/EC.

Ponte a Moriano, April 16, 2013


 "ME. RO" S.p.A.
 Il Presidente
 LAMBERTO MENNUCCI

Signature.....

Use and maintenance manual for generator

MR\VP 20K IGBT N° 130000\C.5110\13 BOBST MANCHESTER



Warning!

This operations manual required for the safe use of the generator. It is therefore advisable to keep it near the unit.

1	16/04/13	First edition		Mario Della Santa	
Rev.	Date	Description	Compilation	Check	Approval

This manual is directed to persons working with the MR\VP generator, in particular to personnel involved in its operation.

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Me.Ro. S.p.A. assumes no responsibility for errors contained in this documentation.

This documentation contains a description of the product that is as accurate as possible.

The product may be subject to technical changes without previous notice.

For any questions or problems regarding installation, contact Me.Ro. S.p.A. We will be pleased to assist you.

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1 GENERAL INFORMATION

*Who is this chapter
directed at?*

Profile of this chapter:

This chapter is directed to persons who install, operate and perform maintenance work on the electronic generator.

Contents of the chapter

This chapter contains the main details of the product and of the manual.

1.1 PRODUCER'S WARRANTY

WARRANTY CONDITIONS:

Me.Ro. S.p.A.'s equipment is built using tested materials. Once manufactured, the equipment is further tested.

12-month warranty

Our equipment is guaranteed **12 months** ex-works our factory, located in Ponte a Moriano, Lucca (Italy). In case of requests for intervention on the equipment during the warranty period, travel hours and costs, food and lodging will be invoiced to the requesting company. Working hours spent to repair the problem and any spare parts necessary are to be borne by *Me.Ro. S.p.A.*.

EXCLUDED FROM WARRANTY:

Damages to the equipment caused by an improper use or by unauthorized operation thereof are excluded from warranty.

1.2

STRUCTURE OF THE DOCUMENTATION

GENERAL

Two types of documents are supplied together with *Me.Ro.spa's* products:

- Use and maintenance manual
This manual contains all details necessary for the unit's installation, operation and maintenance.
- Technical documents concerning the unit
The technical documentation regarding the unit includes layout, drawings and materials list.

This document is the **use and maintenance manual**.

1.3

SYMBOLS LEGEND



Danger!

This signal warns of possible physical injury.



Danger: high voltage

This signal warns of a danger deriving from the presence of high voltage. If not respected, electrical shock may ensue.



Danger: High frequency!

This signal warns of a danger deriving from high frequency radiation. People wearing pacemakers or prostheses should not access the area bearing this symbol. The pacemaker's operation could be disturbed and prostheses' internal temperatures could rise.



Warning!

This symbol calls attention to important information.

1.4

TECHNICAL EXPRESSIONS AND ABBREVIATIONS

DESCRIPTION OF ABBREVIATIONS

<i>P</i>	Power
<i>I</i>	Current
<i>V</i>	Voltage
<i>f</i>	Frequency
<i>L</i>	Inductance
<i>C</i>	Capacity

2 SAFETY

Profile of this chapter:

Who is this chapter directed at?

This chapter is directed to persons who install, operate and perform maintenance work on the electronic generator.

Contents of the chapter

This chapter contains the main details for safe operation of the product.

2.1 INTENDED USE



Danger!

Protection of people and of the system is not guaranteed if the generator does not work in accordance with its intended use.



Any maintenance and repair operations on the electronic generator and on the electrodes must be performed exclusively by authorized specialized personnel.

Plasma applications

INTENDED USE

The generator is conceived as an energy source to supply a vacuum chamber for plasma formation.

The generator is to be used exclusively for its intended use.

2.2 GENERAL WARNINGS

DANGER: PRESENCE OF HIGH VOLTAGE



Danger: high voltage!

*The generator converts network voltage in **DC** and subsequently in the inverter stage in **AC**. Under these voltages, the components can cause electric shock.*



Before intervening on the generator and on the plasma chamber, turn the general switch to OFF.

2.3

DANGEROUS AREAS: MAINTENANCE AND REPAIR

DANGEROUS AREAS: MAIN SWITCH TO OFF



Remove power supply before intervening on the generator.

2.4

SAFETY DEVICES

CONTROL AND PROTECTION DEVICE

Warning!



The operator is responsible for safety operations on the generator. He/she should be acquainted with the norms and apply them.

The electronic generator comes with a general switch and an external emergency button or device.

The external emergency button or device can be connected to the relative terminals located inside the generator (see electrical diagram) by disabling the actuation of the contactors that supply the generator itself.



Danger!

The general switch does not guarantee personal safety and in any case must not be used as an emergency switch.

Before intervening on the generator, wait about a 30" to allow the DC link to discharge.

Protection systems

The generator is immediately turned OFF in case of:

- Overvoltage
- Overheating
- Overloading
- Interlocks
- General switch open

2.5

SAFETY MEASURES



In case of emergency, respect the following steps in the given order:

1. Interrupt the generator's external supply line and any auxiliary supplies.
2. Extinguish the fire through an appropriate extinguisher.



Danger: high voltage!

Due to the presence of high voltage, never extinguish a fire with water but use appropriate extinguishing agents instead.

3 TECHNOLOGICAL DESCRIPTION

<i>Who is this chapter directed at?</i>	Profile of this chapter: This chapter is directed to persons who install, operate and perform maintenance work on the electronic generator.
<i>Contents of the chapter</i>	This chapter contains information on generators' operation and technology, uses and application fields and technical data.

3.1 THE COMPONENTS AND THEIR FUNCTIONS

GENERATOR CONSTRUCTION

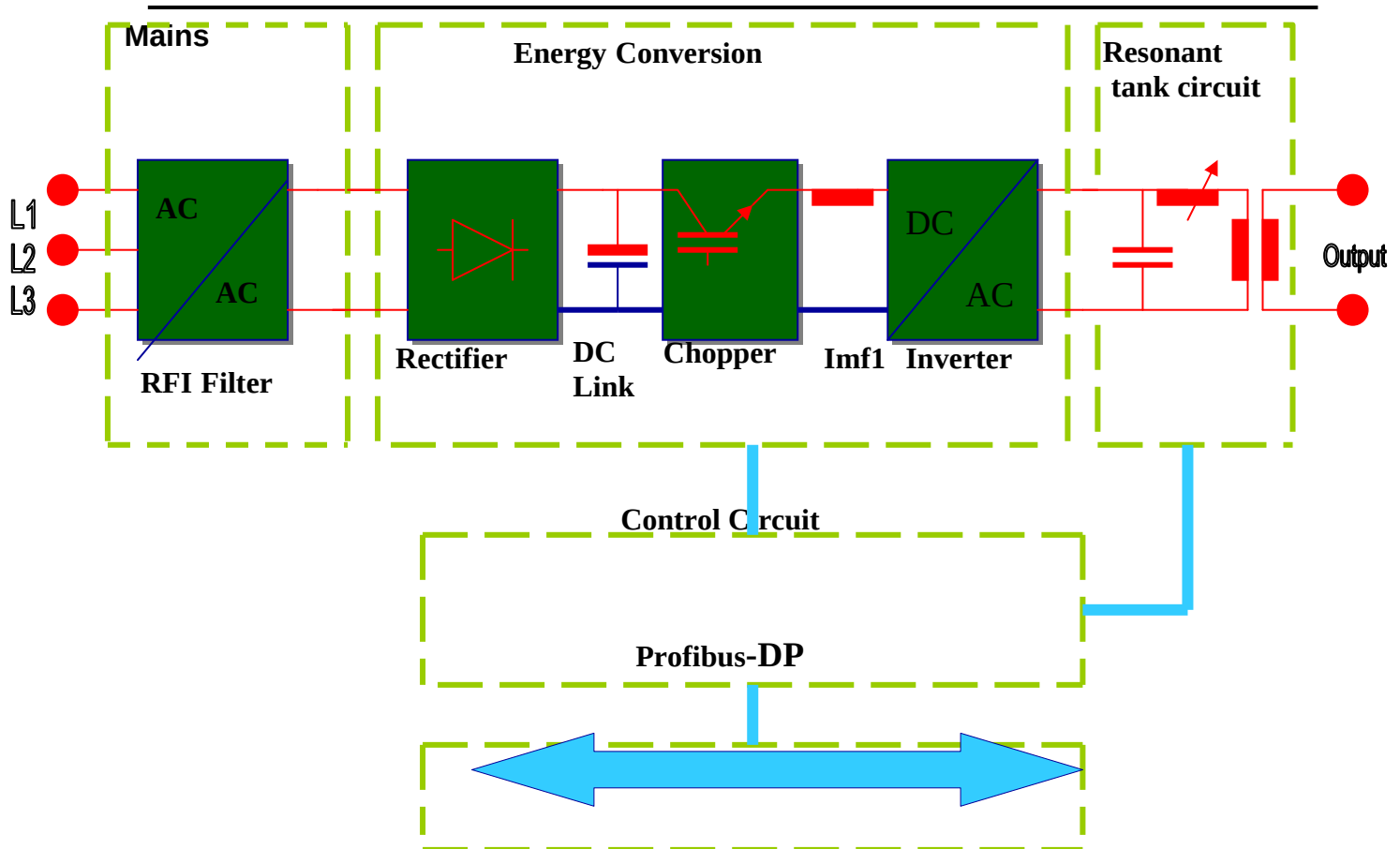
The generator is built in metallic cabinet.

Energy conversion components **IGBT – Diodes – Resonant tank circuit** are air-cooled.

COMPONENTS

The figure below illustrates the generator's diagram in blocks. Essentially, it is comprised of:

- **Mains**
- **Rectifier**
- **Chopper**
- **Inverter**
- **Output transformer**
- **Control electronics**
- **Arcs management**
- **Profibus DP**



Mains: this stage is comprised of an RFI filter that protects the equipment from network interferences and respects the requirements of the norm on electromagnetic compatibility, "EMC", for this type of equipment.

The rectifier converts alternate network voltage into continuous voltage to supply the Chopper stage.

The Chopper regulates the generator's output power, controls current and voltage limitations and, through impedance IMF1, supplies the inverter.

The Inverter is comprised of four IGBT modules in full-bridge configuration, is supplied in current and coupled with a resonant circuit comprised of the output transformer's primary inductance and by the CVOL capacity.

The output transformer, as mentioned above, is an integral part of the inverter; multi-polar secondary winding allows easy adaptation between generator and load.

The control electronics perform all the functions related to the control and piloting of the generator's power stages, as well as the management of the arcs that are created inside the plasma chamber.

Arcs management: when an arc is formed inside the plasma chamber, its load resistance drastically decreases, causing an increase in the output current lout.

If the increase exceeds a determinated set value (value that changes proportionally with the Power Level in the used function mode), the generator is turned to OFF and, after a brief delay, is reset automatically to ON.

When the number of arcs exceeds the set value (*Number of arcs*, *VW2018 Profibus-DP*) in the opening time of the arcs count window (*ArcCountWin.*, *VW2020 Profibus-DP*), a message alarm is displayed: "Arcs limit reached" and the generator is set to OFF. See subsequent table.

If the number of arcs in the count window opening time is less than the set value, the message alarm "Arcs limit reached" is not displayed and the counter is reset upon count window closure

The Profibus-DP allows to remote-control the generator from a supervision device that manages the system.

3.2

CONTROL MODE

PROFIBUS MODULE-DP. EM277

Configuration 6

32 Word Master Input

32 Word Master Output

PROFIBUS-DP BOBST VACUUM PSU – ME.RO Interface table

MASTER	BOBST.	WRITING TO ME.RO	PLC S7 200 SIEMENS CPU 224	SLAVE	ME.RO
PAW X	Bit 0	GENERATOR START-STOP	1 = START 0 = STOP	V2000.0	
	Bit 1	RESET	1 = RESET (pulse)	V2000.1	
	Bit 2	REMOTE CONTROL	1 = REMOTE ENABLED	V2000.2	
	Bit 3	SEL MANUAL	1 = MANUAL MODE SELECTED	V2000.3	
	Bit 4	SEL AUT	1 = AUT MODE SELECTED	V2000.4	
	Bit 5	ARCS RESET	1 = RESET arc counter (pulse)	V2000.5	
PAW X +2		LOW SPEED	LOW SPEED (m\min)	VW2002	
PAW X +4		POWER MAX deviation	MAX. POWER DEVIAT. Allowed %	VW2004	
PAW X +6		CHAMBER LENGTH	PLASMA CHAMBER LENGTH (cm)	VW2006	
PAW X +8		SP MAN.	0 – 100 %	VW2008	
PAW X +10		SP AUT	500 – 10.000 W.	VD2010	
PAW X +14		Spare	Spare	VW2014	
PAW X +16		SPEED	SPEED OF WORK	VW2016	
PAW X +18		NUMBER OF ARCS	SELECT NUMBER OF ARCS	VW2018	
PAW X +20		TIME FOR ARCS	TIME FOR ARCS IN SECONDS	VW2020	
PAW X +22		TAP USED	OUT. TRANSF. TURNS USED	VW2022	
PAW X +24		ARC detection threshold	ARC detection threshold 1:100% 1 = higher sensitivity 100= lower sensitivity	VW2024	

MASTER	BOBST.	READ BY ME.RO	S7 200	SLAVE	ME.RO
PEW X	Bit 0	GENERATOR ON	1 = GENERATOR ACTIVE	V2064.0	
	Bit 1	ALARM 1	1 = ALARM GENERATOR	V2064.1	
	Bit2	ALARM 2	1 = POWER DEVIATION ALARM	V2064.2	
	Bit 3 -15		Spare		
PEW X +2		I. DC	GEN WORKING CURRENT (A.)	VD2066	
PEW X +6		V.DC	GEN WORKING VOLTAGE (V.)	VD2070	
PEW X +10		V.OUT	V. Out rms (Plasma Chamber)	VW2074	
PEW X +12		I.OUT	A. Out rms (Plasma Chamber)	VD2076	
PEW X +16		P.OUT	GENERATOR POWER OUTPUT (W.)	VD2080	
PEW X +20		Spare	Spare	VD2084	
PEW X +24		ALARM	WORD ALARM	VW2088	
PEW X +26		SIGNALLING	WORD SIGNALLING	VW2090	
PEW X +28		ARCS	NUMBER OF ARCS	VW2092	
PEW X +30		SPEED	LINE SPEED m/min	VW2094	

WORD ALARM - VW2088 -

Word VW2088 assumes different values based on the alarm:

VW2088 = 0 Ready to start

VW2088 = 1 Low Speed

VW2088 = 2 Emergency

VW2088 = 4 Vcc Fault

VW2088 = 8 Thermal overload

VW2088 = 16 Cooling water plasma chamber

VW2088 = 32 Metallizer open

VW2088 = 64 Overload

VW2088 = 512 Arc limit reached

WORD SIGNALLING – VW2090

Word VW2090 assumes different values according to the signalling to be made:

VW2090 = 1 Ready to start

VW2090 = 4 Generator ON

VW2090 = 8 Low speed

VW2090 = 16 Present alarm

VW2090 = 32 Power too high

VW2090 = 64 Power too low

4 INSTALLATION

Profile of this chapter:

This chapter is directed to persons who install the electronic generator.

Who is this chapter

directed at?

Contents of the chapter

This chapter contains information concerning the proper mechanical and electronic installation of the generator and the station.

4.1 TRANSPORT, PLACEMENT

TRANSPORT

Once you receive the equipment:



Check the integrity of all its parts so as to verify any damage that may have been caused by transport.



Immediately signal any damage that may have been caused by transport.

For shipment, the system's components are packed in appropriate containers having a wooden base.



We advise that you remove the packaging only when the machine is near to its definitive position.

The machine may be moved with a forklift of suitable capacity.

The gross weight is reported on the packaging.

To move the machine without its packaging, both the generator and the discharge stations must be lifted using pre-arranged ring-bolts.

PLACEMENT

*Environmental
conditions*

The machine must be adequately protected against water and dust.

It may be warehoused at temperatures included between -25°C and +70°C.

A dry environment is preferable (not mandatory).

4.2 UNPACKING



Carefully unpack the various components of the generator

The packing contains all the cables necessary to start the machine's operation, except for the main supply line.

4.3

POSITIONING

When installing the machine, possible dangers that may arise from uncontrolled moving of the machine have to be kept in consideration. In these situations, it is necessary to block the machine to the floor.

SPACE NECESSARY FOR USE AND MAINTENANCE

For check up and maintenance of the generator it is necessary to have the possibility to remove the front and the top panel.

Pay particular attention to air outlets for generator cooling; these outlets must be free and easily accessible for maintenance (filter cleaning etc.).

ENVIRONMENTAL CONDITIONS ALLOWED

Environmental working conditions

The electronic generator can work at temperatures included between +5°C and +40°C, humidity levels of 15%-85% without condensation.

4.4

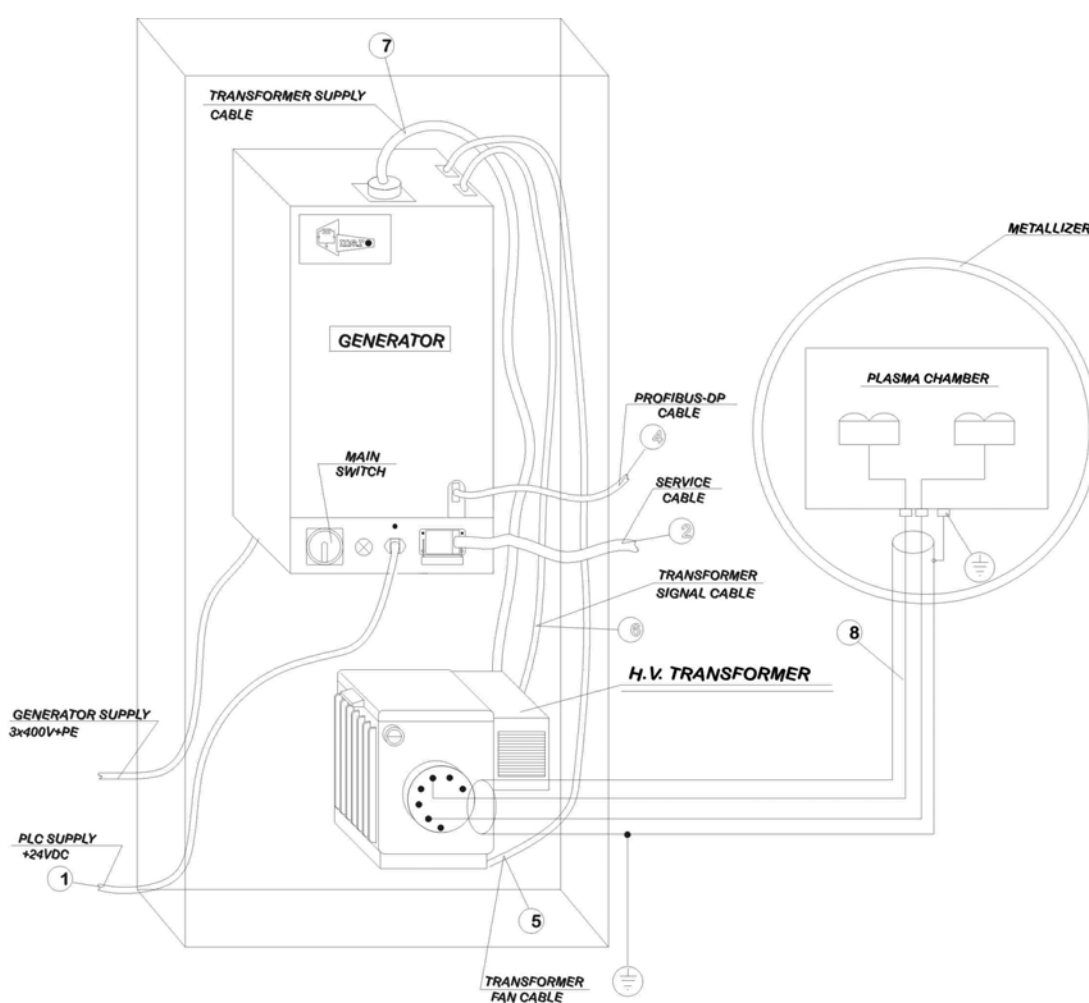
CONNECTIONS



Danger: high voltage!

The generator's electrical connections must be performed by qualified personnel, trained to work with electronic devices.

These persons must follow local safety rules for the installation of the unit.



OUT OF THE SYSTEM

LAY

Supply line size

POWER LINE

Protections on the general supply panel (switch plus fuses) must be of the appropriate size, according to the supply voltage and the max. absorbed current by the generator; see INPUT electrical data label located on the back of the generator's cabinet.

Specifications

ELECTRONIC GENERATOR "Solid State" MR/VP 20K IGBT

SUPPLY VOLTAGE: **3x380-480 V* – 50/60 Hz**

***The voltage of the TR1 services transformer inside the generator must be the same as the supply voltage**

MAX. ABSORBED POWER: 22,5 kVA

MAX. OUTPUT POWER: 20 kW

WORK FREQUENCY: 35 ÷ 45 kHz

AIR COOLING

DIMENSIONS WxHxD = 600 x 2000 x 600 (mm)

WEIGHT kg. 220

TRANSFORMER MR\T-VP 20K CH

DIMENSION WxHxD = 580 x 440 x 498 (mm)

WEIGHT kg. 98

4.5

INSTRUCTIONS FOR SETTING UP THE GENERATOR

Type MR/VP 20K IGBT (see connections drawing)

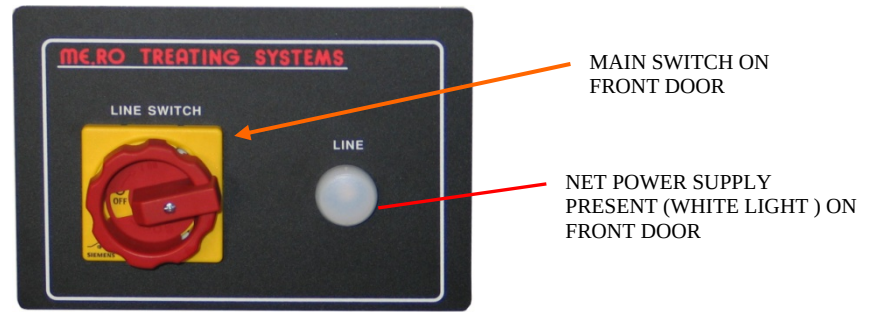


Fig.0 "Panel on front door"

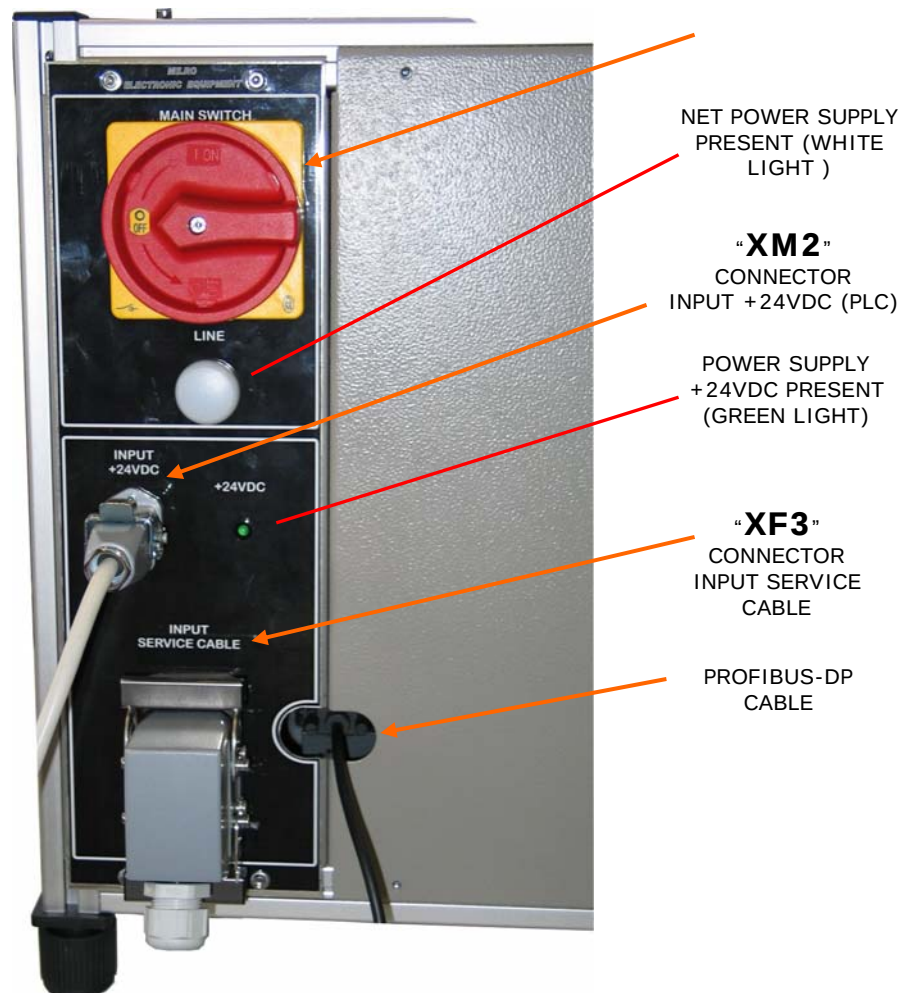


Fig.1 "Generator Panel"



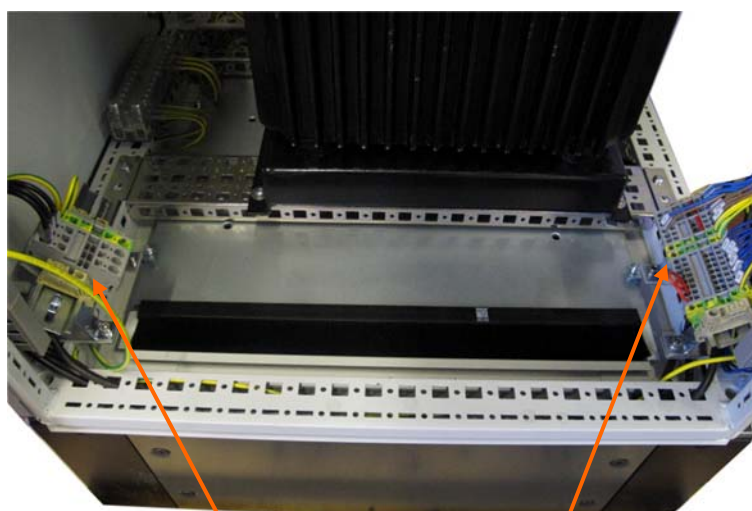
"IA1"
MAGNETOTHERMIC
SWITCH.

**SWITCH ONTO
POSITION I (UP ↑).
ONLY WHEN THE
MAIN SWITCH IS IN
OFF-POSITION**

"X0" SOCKET
" INPUT LINE

see INPUT data label
located on the
generator's cabinet

Fig.2 " Generator Supply Line"



"X0" SOCKET
INPUT MAIN LINE

"X5" SOCKET
INPUT COSTUMER
INTERLOCKS

GENERATOR POWER SUPPLY

Following the electrical diagram, connect the line cable to the socket **X0** fitted in to the chassis as in fig.2



N.B. The generator supply line must be protected by a differential switch to interrupt supply in case of loss of isolation.

The cable must be of appropriate size, keeping into account the max absorbed current (see INPUT data label located on the generator's cabinet).

PLC S7-200 SIEMENS POWER SUPPLY

Connect the connector **XF2** to the connector **XM2** fitted on the front panel of the chassis as in fig.1 (**SUPPLY +24VDC**) (for PLC).

SERVICES AND SAFETYS

Connect the connector **XM3** to the connector **XF3** fitted on the front panel of the chassis as in fig.1 (**INPUT SERVICE CABLE**)

OUTPUT TRANSFORMER SIGNALS

Connect the cable "SIGNAL CABLE" between connector **XM4** and the **XF4** on output transformer as in fig. 3.

POWER SUPPLY OF THE OUTPUT TRANSFORMER FAN

Connect the cable "SUPPLY TRANSFORMER FAN CABLE" between connector **XF5** and the **XM5** on output transformer as in fig. 3 and fig.4.

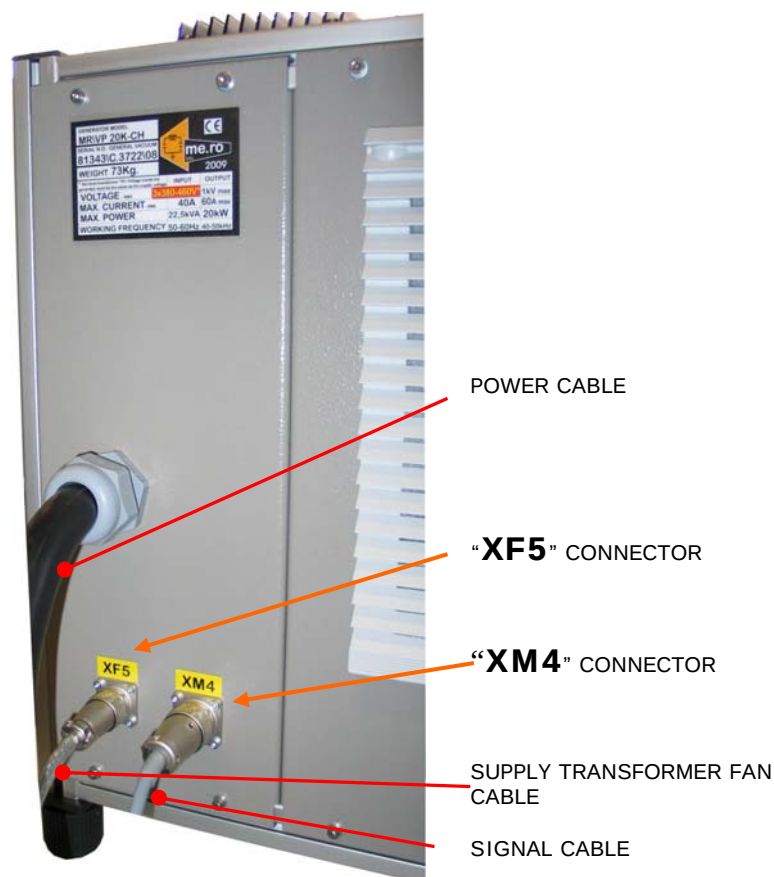


Fig. 3 "Generator output lines"

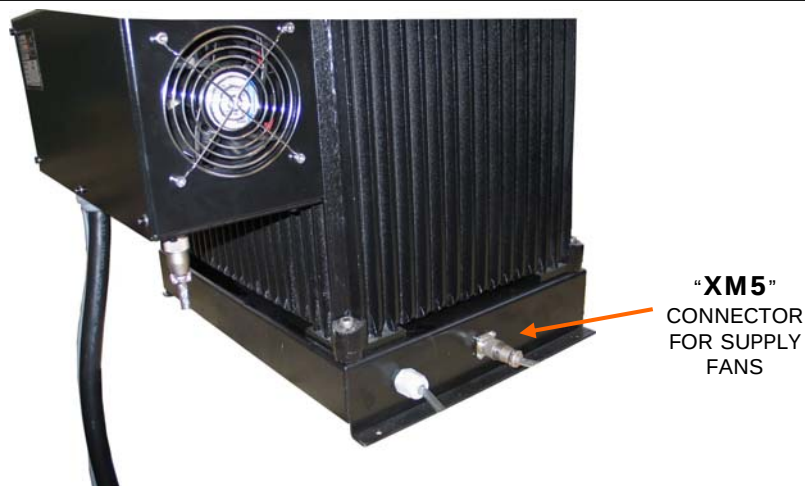


Fig.4 "Outside Transformer"

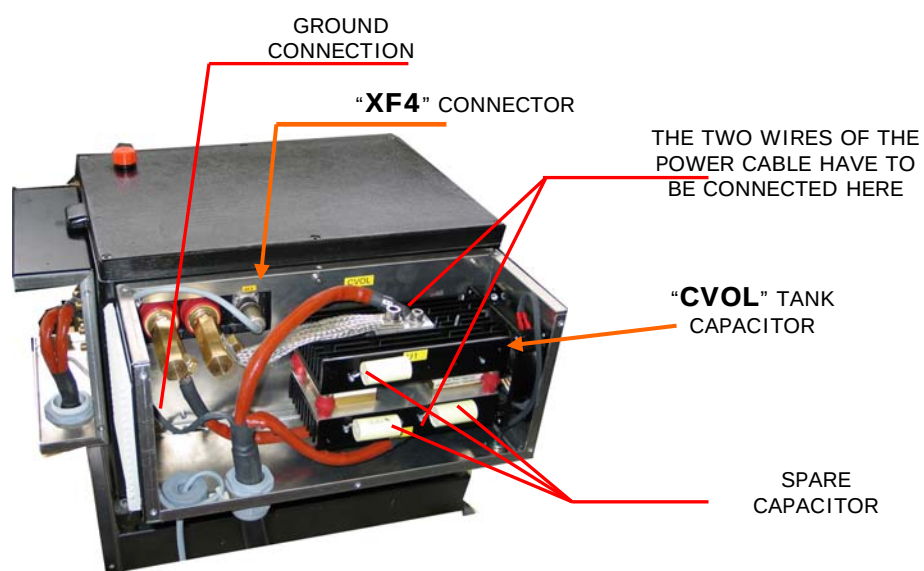


Fig.5 "Inside box Transformer"

➤ THE TWO WIRES OF THE POWER CABLE COMING FROM THE GENERATOR HAVE TO BE CONNECTED TO THE JUNCTION BOX OF THE OUTPUT TRANSFORMER (fig.5).

➤ THE GROUND WIRE HAS TO BE CONNECTED TO THE APPPOSITE SCREW INSIDE THE JUNCTION BOX OF THE OUTPUT TRANSFORMER (fig.5).



DO NOT TURN ON THE GENERATOR BEFORE HAVING CONNECTED THE POWER SUPPLY CABLE TO THE OUTPUT TRANSFORMER .

4.6

ADAPTATION OF THE GENERATOR TO THE LOAD

Secondary taps of the output transformer

MODEL MRIT-VP 20K-CH			
SERIAL N.O. 81344\C.3722\08		CE 2009	
WEIGHT 98 Kg.			
OUTPUT POWER		20kW	
WORKING FREQUENCY		40-50kHz	
SEC. WINDING TAPS	1	2	3 4 5 6 7 T.T. 200V
SEC. WINDING TURNS	5	6	7 8 9 10 11 TURNS
SEC. WINDING INDUCT.	25,1	35,0	46,4 59,3 73,6 89,5 107 uH



Fig.6 "Output Trasformer Taps"

GROUND
CONNECTION
FROM THE PLASMA
CHAMBER

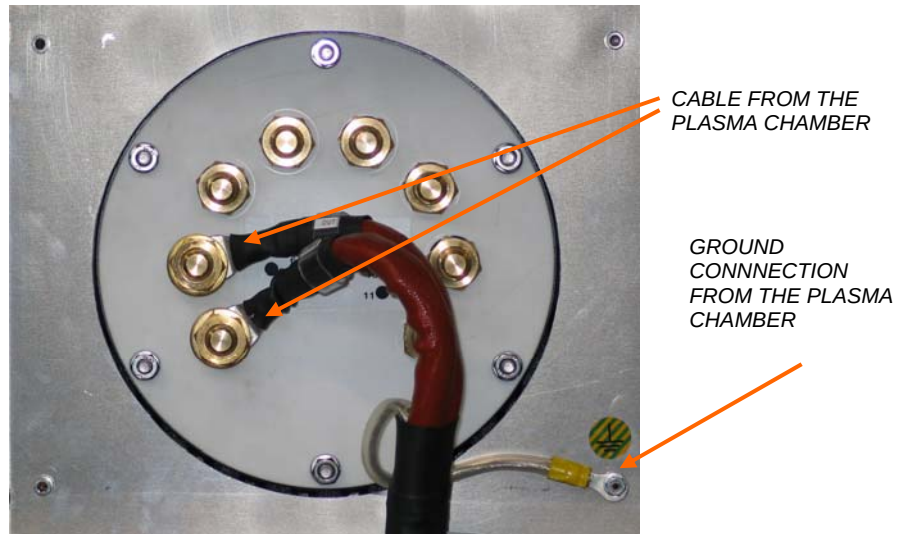


Fig.7 "Output Trasformer Taps with cable"

The data label on the output transformer indicates the taps and the relevant turn number. The turn number of each tap is also indicated on the white label put on the white insulator.

- Before turning the generator ON, **check that the vacuum conditions and the gas issued in the plasma chamber are those of standard operation.**
- Select the **manual** operation mode.
- Bring the setpoint for power adjustment to zero and turn the generator ON.
- Read the output values **PW** (VD2080), **IDC** (VD2066), **VDC** (VD2070) **PROFIBUS-DP** on the supervisor. Consider that the max. output power of the generator with the setpoint for power adjustment equal to **100 is 20kW**, and the values **IDC and VDC are respectively 50A. 400V.**
- Gradually increment the power adjustment setpoint up to 100 and check the generator's output power. If the generator is not correctly adapted to the load, it will not be able to issue the max power and in this case, two situations may arise:

1° The value of the VDC. reaches 400V. but the IDC current remains below 50A.

2° The value of the IDC reaches 50A. But the VDC voltage remains below 400V.

In the first case, it is necessary to utilize a higher numbers of turns on the output transformer's secondary.

In this way, the VDC value decreases while the IDC value increases.

In the second case, it is necessary to use a lower tap on the output transformer's secondary.

In this way, the VDC value increases while the IDC value decreases.

If max output power is not required, the generator can correctly work also with IDC VDC values different from the max limit of both.

The max value of the individual sizes is in any case limited for the VDC to 400V and for the IDC to 50A.

If necessary, in order to have a fine adjustment, it is possible to modify the variable impedance which is serial connected to the primary coil of the output transformer (see electric diagram of the generator). The impedance is fitted inside the output transformer box.

May be varied by adding or removing the 6 ferrites which are inside the impedance (see fig. 8-11).

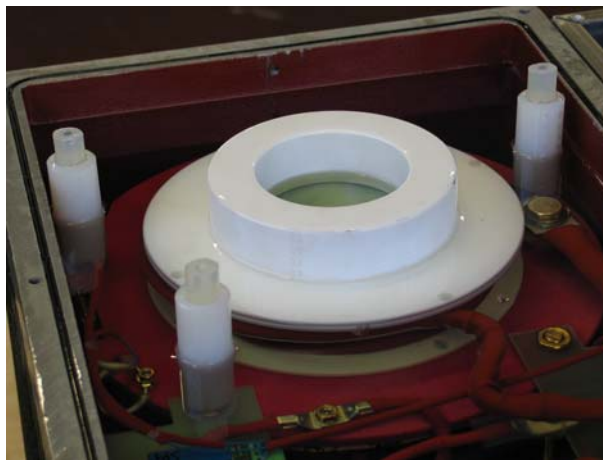


Fig.8

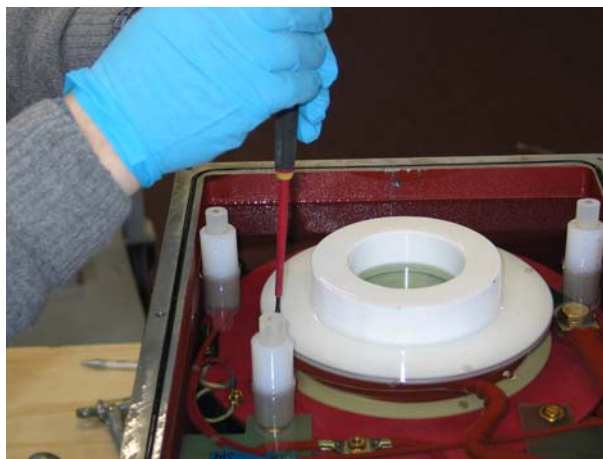


Fig.9

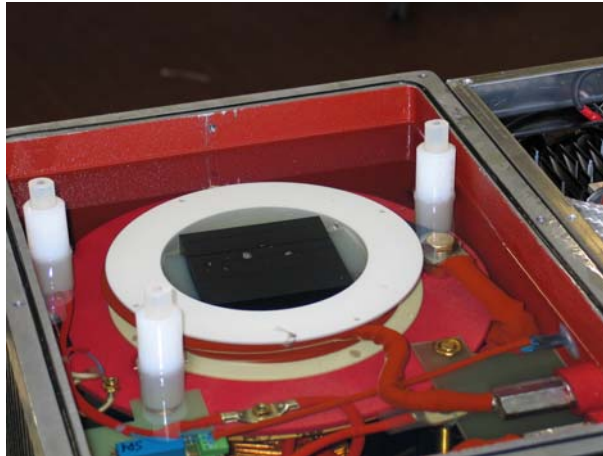


Fig.10

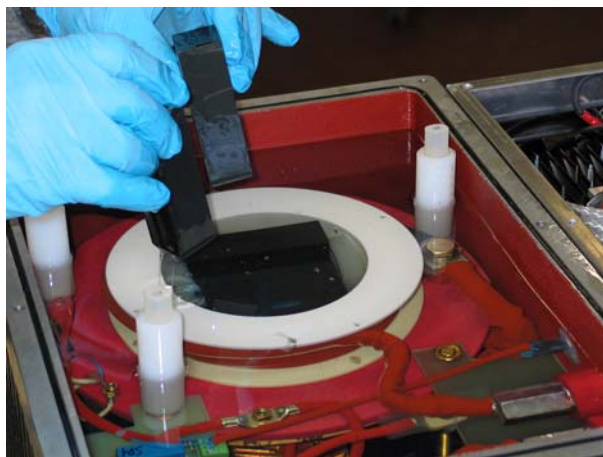


Fig.11

Summarizing, optimal working conditions for the generator with max output power are:

PW = 20 kW

VDC = 400 V.

IDC = 50 A.

These values are obtained by using the most appropriate tap on the output transformer's secondary so as to obtain optimal match between generator and load.

If the plasma chamber is not able to work with the generator's max output power, or if the metallization process does not require max power, the generator can work without any problems (as already mentioned above) even if optimal matching conditions are not reached

5 OPERATION, CONTROL

Who is this chapter directed at?

Profile of this chapter:

This chapter is directed to persons who utilize the electronic generator.

Contents of the chapter

This chapter contains a description of the generator's operational modes and states.

5.1 SETTING UP

The generator has been tested by *Me.Ro. S.p.A* at its premises.

It can be set up by following what explained in paragraphs 3.2 4.4 4.5

5.2 STRUCTURE OF THE OPERATOR TERMINAL

In this specific application, the operator terminal is replaced by specific pages of the supervision system. Therefore, to set the variables and the commands, the operator must refer to the specific instructions of the supervisor.

5.3 TURNING THE GENERATOR ON/OFF

CHECKING SAFETY DEVICES



Before turning the generator on, check the following points:

- All safety elements must be active
- All remote interlocks must be closed

PREPARATION FOR OPERATION



Supplying the generator.

Checking the following points:

- Check the alarm signal menu on the operator terminal.
- Remove any alarms present

NB. The generator may be turned on only after having eliminated all alarms.

5.4 START OF OPERATION

The generator is turned ON by the supervision program.

5.5 TURNING THE GENERATOR OFF

The generator is turned OFF by the supervision program.

5.6 TURNING OFF IN CASE OF EMERGENCY



Press the emergency button.



Danger: high voltage!

After having pressed the emergency button, the generator may have residual voltages (see chapter 2: "Safety")

5.7 SETTING PARAMETERS



Before turning the system on, you must set some parameters on the Operator Terminal (supervision) using the following page.

Remote Control. Bit Profibus – DP enabling ****

REMOTE CONTROL	1 = REMOTE ENABLED	V2000.2
----------------	--------------------	---------

Operation mode. Bit Profibus – DP selection

SEL MAN.	1 = MANUAL MODE SELECTED	V2000.3
SEL AUT.	1 = AUTOMATIC MODE SELECTED	V2000.4

Arc Reset. When the number of arcs in the counter window overtakes the one set in VW2018 set the BIT V2000.5 to one to reset the arc counter

ARC RESET	1 = RESET arc counter (pulse)	VW2005
-----------	-------------------------------	--------

Low Speed. Setting the minimum speed below which the generator turns to OFF. Word Profibus – DP setting

LOW SPEED	LOW SPEED (m/min)	VW2002
-----------	-------------------	--------

Power max. deviation. Setting the max output power deviation % admissible before enabling the signalling “Power too High” or “Power too Low” (see message and signalling alarms paragraph 5.9). Word Profibus – DP setting

POWER MAX DEVIATION	MAX POWER DEVIAT. Allowed %	VW2004
---------------------	-----------------------------	--------

Plasma chamber length. Setting plasma chamber length. (This datum, together with speed, is used to calculate production in m²/min.) Word Profibus – DP setting

PLASMA CHAMBER LENGTH	PLASMA CHAMBER LENGTH cm.	VW2006
-----------------------	------------------------------	--------

Power control. Setting the Setpoint to adjust output power. Word Profibus – DP setting

SP MAN	MANUAL SETPOINT 0 100 %	VW2008
SP AUT	AUTOMATIC SETPOINT 500 10.000 W	VD2010

Speed. Speed of work

SPEED	SPEED OF WORK (number)	VW2016
-------	------------------------	--------

Number of arcs. Setting the number of admissible arcs in the opening time of the count window (**See Arc Management paragraph 3.1**). Word Profibus – DP setting

NUMBER OF ARCS	SELECT NUMBER OF ARCS	VW2018
----------------	-----------------------	--------

Time for arcs. Setting time in seconds for the arc count window. (**See Arc Management paragraph 3.1**). Word Profibus – DP setting

TIME FOR ARCS	TIME FOR ARCS IN SECONDS	VW2020
---------------	--------------------------	--------

Tap used. Setting the secondary turns used on the output transformer (this datum is used to calibrate the reading of the Vout rms). Word Profibus – DP setting

TAP USED	OUT.TRANSF. TURNS USED	VW2022
----------	------------------------	--------

Arc sensitivity. Set a number between 1 to 100 to changed the arc threshold. Default will be 30 as a starting point.

ARC detection threshold	ARC detection threshold 1:100% 1 = higher sensitivity 100= lower sensitivity	VW2024
-------------------------	--	--------

5.8 OPERATION MODE

Power control on the generator can be performed in three different ways.

Manual mode.

Automatic mode W

Selection is made by acting on bits 3 - 4 - of V2000 (Profibus-DP).

MANUAL MODE	1 = SEL. MANUAL	V2000.3
AUTOMATIC MODE W	1 = SEL. AUT.	V2000.4

Manual mode

In the manual mode, the generator's output power can be regulated from minimum to maximum through the value "**SP MAN 0 100 %**" inserted in the Word **VW2008** (Profibus-DP)

Automatic mode

In the automatic mode of operation, output power is regulated by setting the setpoint with the desired output power value. "**SP AUT. 1000 10000 W**" inserted in word **VD2010** (Profibus – DP)

By using this type of operation, any load deviations due for example, to a variation in the vacuum degree inside the chamber, or to a change in the quantity of gas issued with consequent change in the output power, are automatically compensated for.

5.9

ALARM AND SIGNAL MESSAGES

Alarm messages

Alarm messages are the following:

- **LOW SPEED:** Production speed inferior to the speed set in the Set Parameters page on the operator terminal.
- **EMERGENCY:** Emergency button pressed.
- **OVERLOAD:** Intervention of peak current limitation due to a generator overload or to having exceeded the number of arcs with respect to the number set on the Set Parameters page of the operator terminal.
- **Vcc FAULT:** Lack of continuous supply voltage at the Chopper stage.
- **THERMAL OVERLOAD:** Intervention of the generator's thermal protections.
- **COOLING WATER PLASMA CHAMBER:** Lack of cooling liquid in the plasma chamber.
- **ARC LIMIT REACHED:** the arc number has exceeded the value which has been set in word 2018 .
- **METALLIZER OPEN:** Metallizer safety interlock is open

SIGNAL MESSAGES

- **LOW SPEED:** Production speed inferior to the speed set on the Set Parameters page of the operator panel.
- **READY TO START:** No alarm in the generator is ready for ON.
- **PRESENT ALARM:** An alarm that is keeping the generator OFF is present.
- **GENERATOR ON:** Generator in operation.
- **POWER TOO HIGH:** Increase in the output power beyond the preset limit % (Admissible Power Dev.) in the Set Parameters page on the Operator Panel.
- **POWER TOO LOW:** Decrease in the output power beyond the preset limit % (Admissible Power Dev.) in the Set Parameters page on the Operator Panel.

The above messages are represented by the absolute value assumed by the **Word Alarm VW2084** and by the **Word Signalling VW2086**.

See paragraph 3.2 Profibus – DP interface table

The work setpoint deviation causes the intervention of the “Power too High” or “Power too Low” alarm signalling, without blocking the generator. The signalling must be restored by supplying an impulse on the V2000.1 (see Profibus DP table).

6 MAINTENANCE

Who is this chapter directed at?

Profile of this chapter:

This chapter is directed to persons perform maintenance work on the electronic generator.

Contents of the chapter

This chapter contains advice and notices regarding the generator's maintenance.

6.1 GENERAL NOTES ON MAINTENANCE

In this charter you will find maintenance advice that describes the work necessary in order to maintain the generator's proper operation through time.

Extraordinary maintenance should be performed by *Me.Ro. S.p.A.* 's technical personnel.

TECHNICAL SUPPORT

Me.Ro. S.p.A.'s technical personnel can be contacted via phone at the company's office numbers.

6.2 SAFE MAINTENANCE



To protect one's own safety, we advise that you follow the safety rules and measures described here following each time the generator requires maintenance.

DANGEROUS AREAS



Danger: high voltage!

The generator converts low frequencies in DC energy. Particulars under high voltage may cause electrical shock.



Ensure that you are well acquainted with the generator's dangerous areas. Carefully read chapter 2.3 "Dangerous areas: Maintenance and repair"

REMOVAL OF RESIDUAL VOLTAGE

Danger: high voltage!



After having turned the generator off, some parts may present residual voltage (see chapter 2: "Safety")



Disconnect the generator and follow the safety norms concerning high electrical voltage systems.

6.3

MAINTENANCE INTERVALS

Regular maintenance

Regularly inspect the RF generator so as to be able to operate it for long periods of time.

Maintenance work should be performed keeping in mind the working time and the environmental conditions under which the generator operates.

6.4

DISMANTLING AND PACKING

UNIT ISOLATION

1. Turn the generator off
2. Interrupt the generator's supply line
3. Remove all external electrical connections

Remove all connections for water flow and completely empty the cooling circuit. Remove all the water from the circuit also with the aid of compressed air.



Danger!

Make the cooling water flow off in the opposite direction with respect to the normal working flow

PACKING



Pack the generator according to its dimensions.

Place it on a pallet, cover it with plastic film and wrap it with belts.



TRANSPORT

See chapter 5.1 "Transport"

DISPOSAL

The generator is composed mainly by.

Metals:

- a. Aluminium
- b. Copper
- c. Steel
- d. Brass
- e. Galvanized iron sheets
- f. Steel sheets

Plastics:

- g. Teflon , PA and others

Other materials:

- h. Electronic cards
- i. Ceramic

Cyclostyled sheet owner in April 13

Me.Ro S.p.A. © 2009 mail - info@mero.it http - www.mero.it

Fig. C

Prese d'aria di raffreddamento
Cooling air intakes

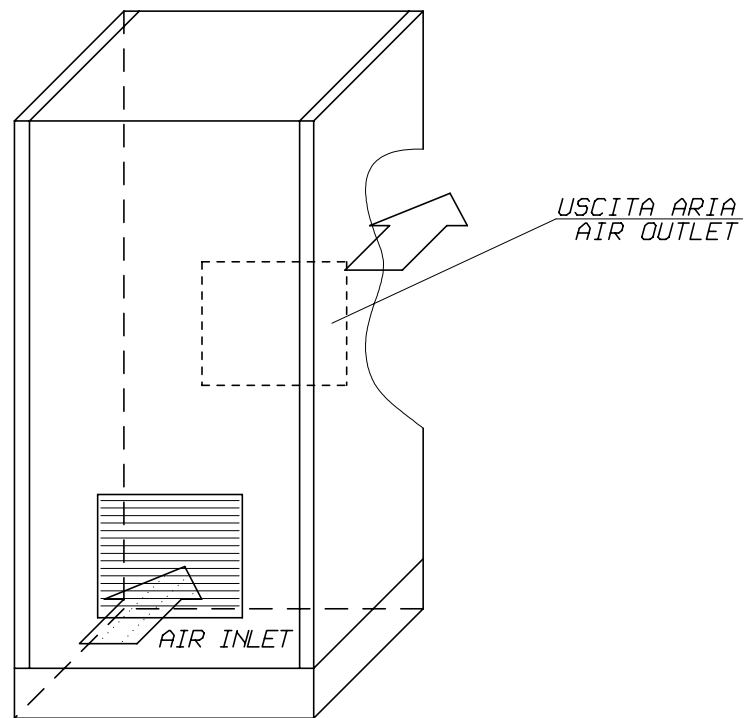


Fig. D

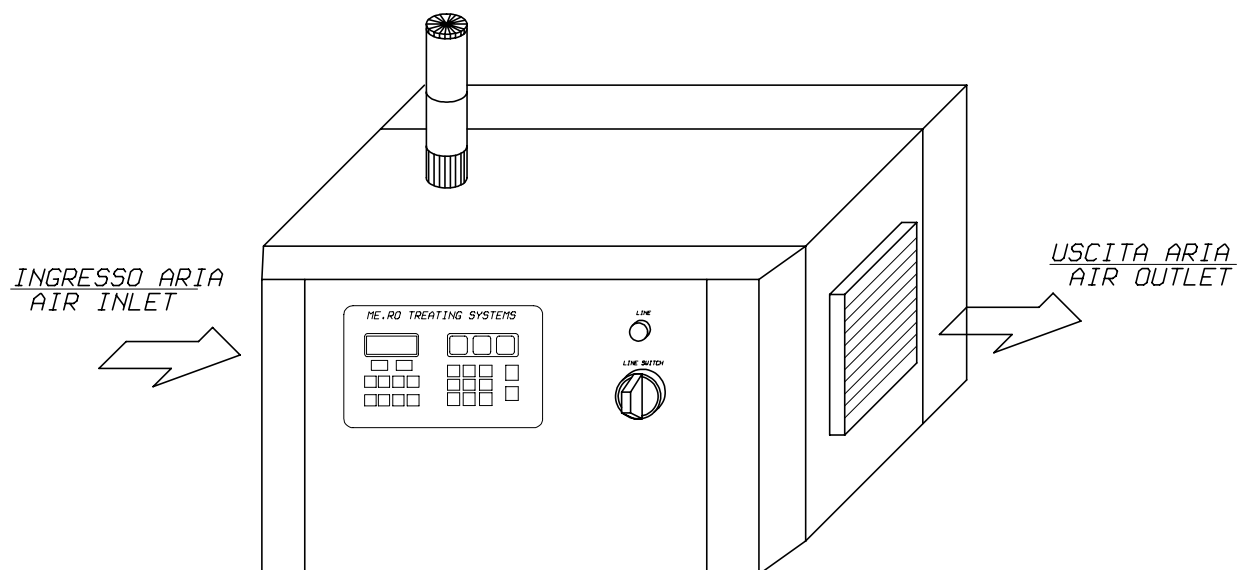
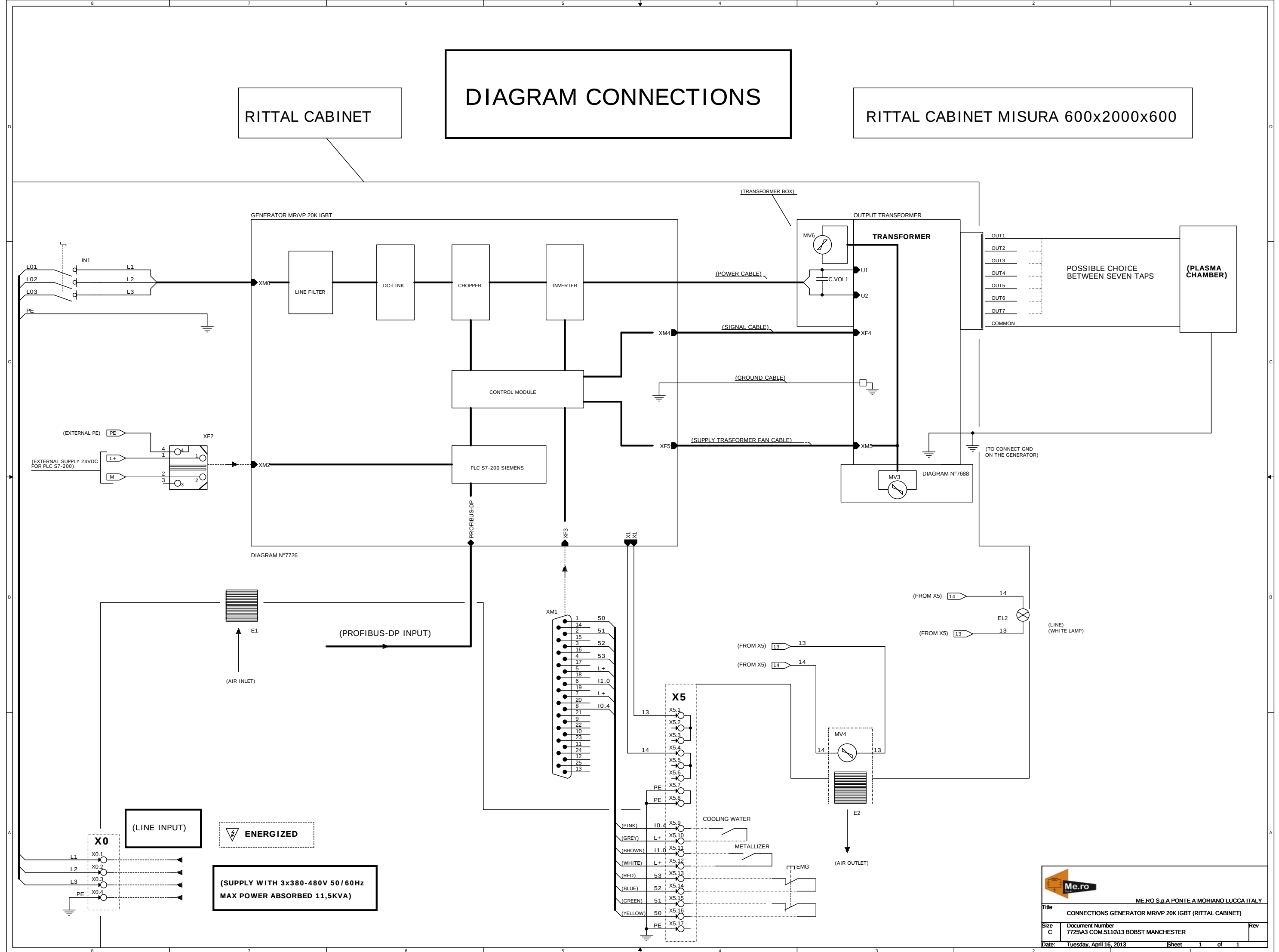
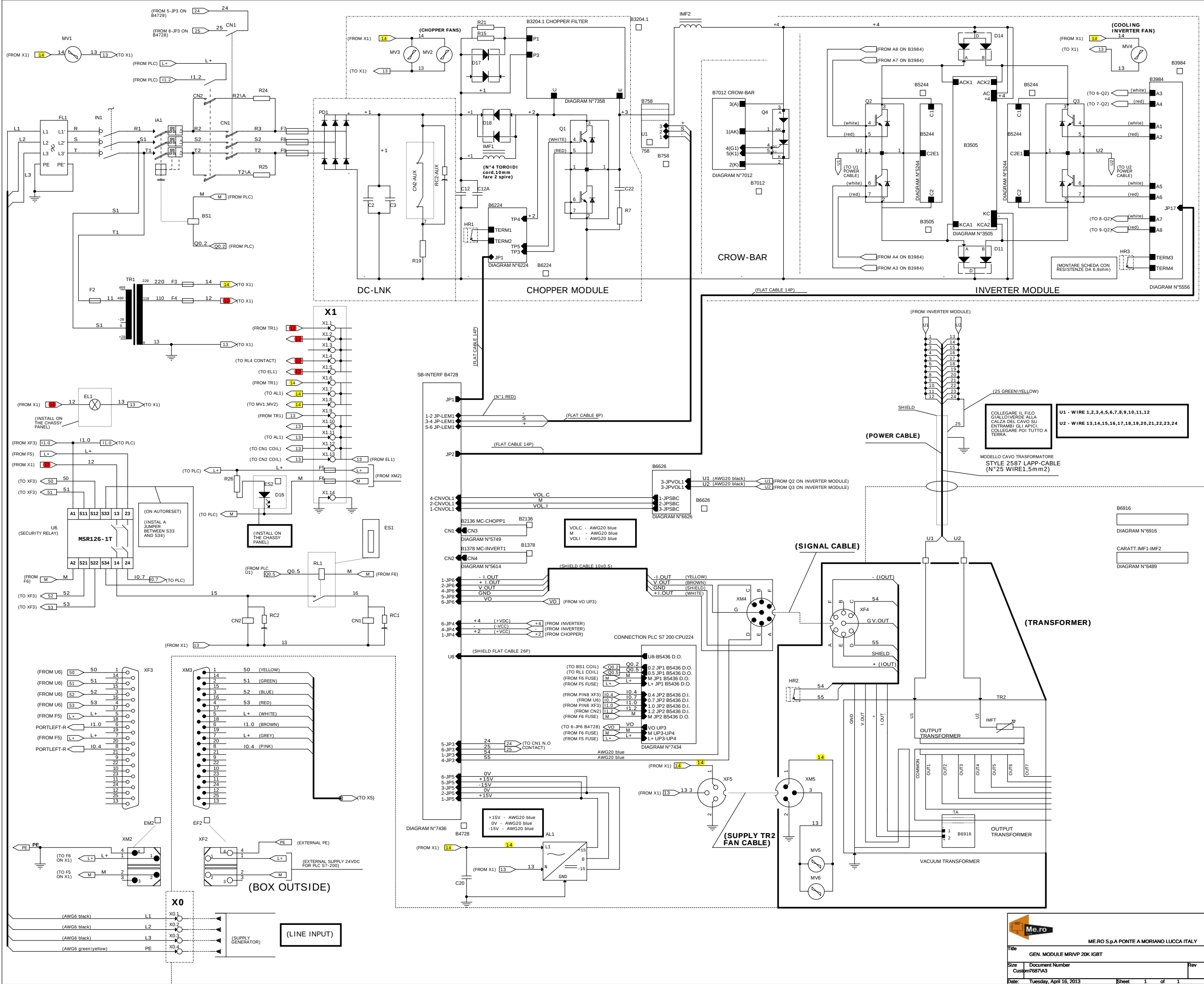


DIAGRAM CONNECTIONS

RITTAL CABINET

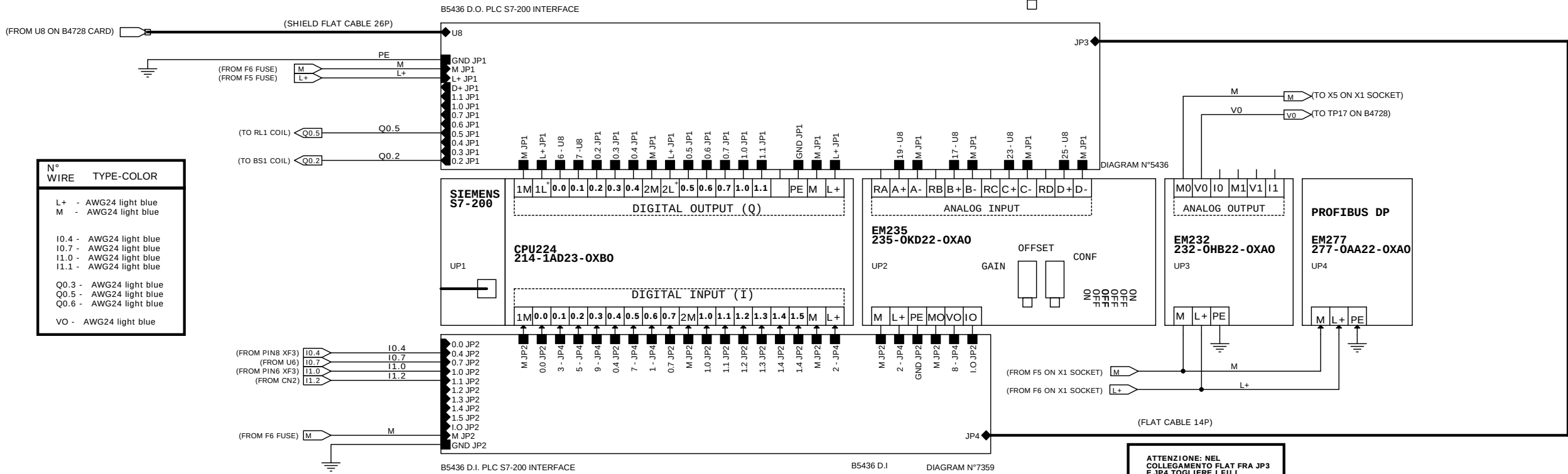
RITTAL CABINET MISURA 600x2000x600



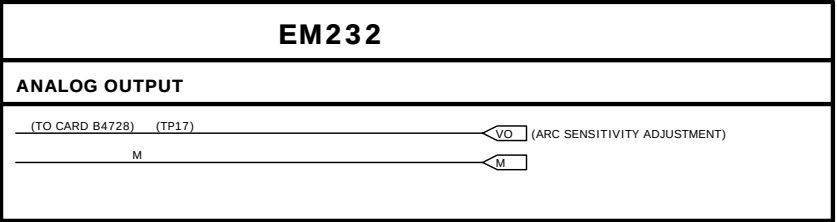
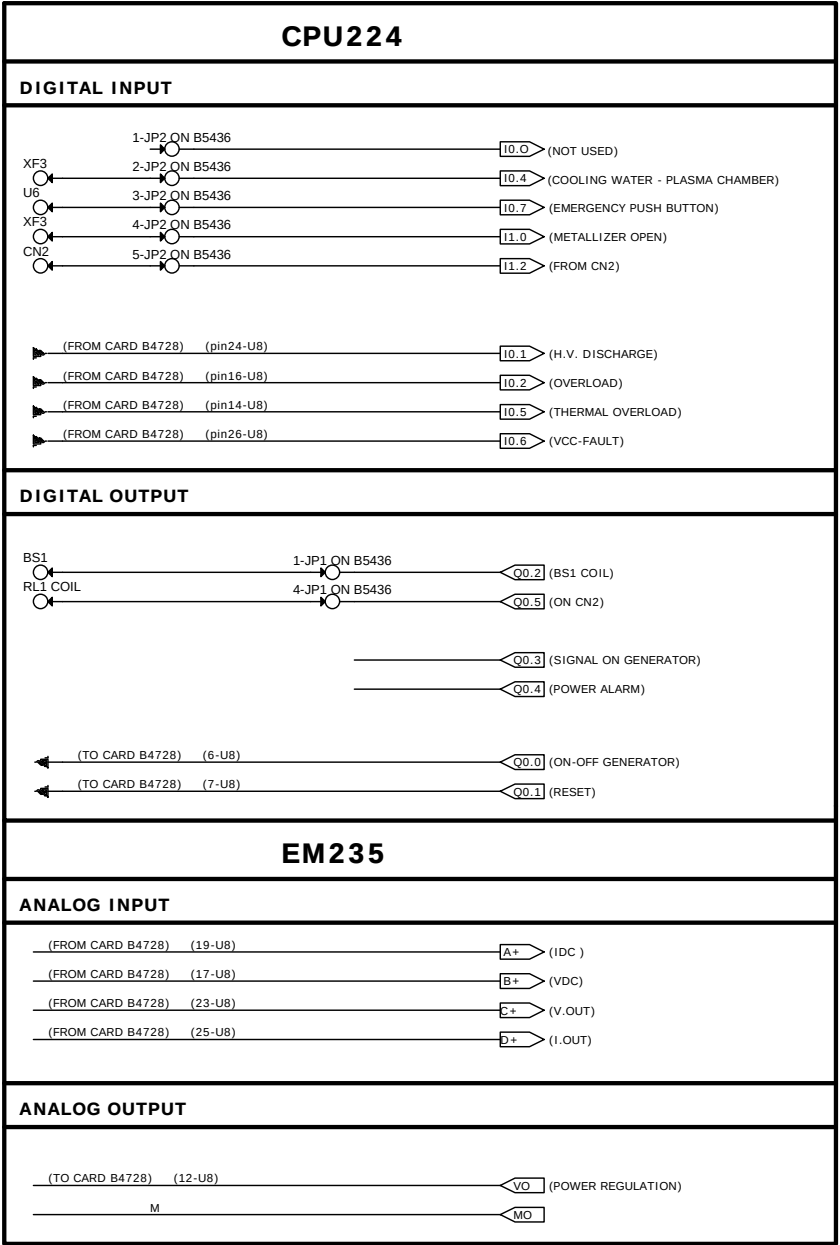


ME.RO S.p.A PONTE A MARIANO LUCCA ITALY

Title		GEN. MODULE MR/VP 20K IGBT
Size	Document Number	Rev
Custom	7687VA3	
Date:	Tuesday, April 16, 2013	Sheet 1 of 1



INPUT-OUTPUT LOGICAL PLC



CONNECTIONS GENERATOR MR/VP 20K IGBT (RITTAL CABINET) Revised: Tuesday, April 16, 2013

7725\A3 COM.5110\13 BOBST MANCHESTER Revision: ?

ME.RO S.p.A PONTE A MORIANO LUCCA ITALY

55029-VIA BALESTRERI N.430

TEL.0583-406060

Bill Of Materials April 16,2013 17:59:04

N°	Quantity	Diagram Reference	Component Type	Component Model	Me.Ro Code
1	1	C.VOL1	TANK CAPACITOR	2//0,66uF MOD. CFM 200 700Vrms	*****
2	1	EL2	SIGNAL LIGHT	M22-LC-W MOELLER	10488
3	1	E1	GRILL	AIR FILTER SK3325-207 255X255	16417
4	1	E2	GRILL	AIR FILTER SK3322-207 125x125	16241
5	1	IN1	MAIN SWITCH	P1-32/V/SVB-ZFS68 MOELLER	84305
6	1	MV4	FAN+AIR FILTER	SK3243-100 230VAC RITTAL	*****
7	1	XM1	CONNECTOR DB25 F	FLAT CONN. 25P MA	42195
8	3	X0.1,X0.2,X0.3	SPRING CLIP	ST-6 PHOENIX	196911
9	1	X0.4	EARTH SPRING CLIP	ST 6PE PHOENIX	196211
10	14	X5.1,X5.2,X5.3,X5.4,X5.5,X5.6,X5.9, X5.10,X5.11,X5.12,X5.13,X5.14,X5.15,X5.16	SPRING CLIP	ST-2,5 GREY PHOENIX	196748
11	3	X5.7,X5.8,X5.17	EARTH SPRING CLIP	ST-2,5/PE PHOENIX	196711
12	1	AL1	POWER SUPPLY	MEAN WELL T40-C	15400
13	1	BS1	RELEASING MECHANISM LAUNCH	FAZ XAA-C 12-110VAC MOELLER	*****
14	1	B758	ELECTRONIC CARD	B758 LEM CIRCUIT	200945 EST
15	1	B1378	ELECTRONIC CARD	INVERTER CONTROL CARD B1378	200164 EST
16	1	B2136	ELECTRONIC CARD	CHOPPER CONTROL CARD B2136	200450 EST
17	1	B3505	ELECTRONIC CARD	INVERTER FILTER B3505	*****
18	1	B3984	ELECTRONIC CARD	B3984 INVERTER PILOTAGE MODULE (RES. 6,8ohm)	2004395 EST
19	1	B4728	ELECTRONIC CARD	MOTHER BOARD B4728	146998
20	1	B5244	ELECTRONIC CARD	INVERTER IGBT FILTER B5244	*****

21	1	B6224	ELECTRONIC CARD	B2561 CHOPPER PILOTAGE MODULE	200945 EST
22	1	B6626	ELECTRONIC CARD	B6626 REACTION BOARD	200626 EST
23	1	B7012	ELECTRONIC CARD	B7012 CROW-BAR	*****
24	1	B3204.1	ELECTRONIC CARD	CHOPPER IGBT FILTER B3204.1	*****
25	1	CN1	CONTACTOR	DILM32-10 110VAC MOELLER	
26	1	CN2	CONTACTOR	DILEM-10 110VAC MOELLER	198775
27	1	CN2-AUX	AUX CONTACT	02DILEM 2NC	198733
28	2	C2,C3	CAPACITOR	C-44U 500uF 900VDC DUCATI	36595
29	2	C12A,C12	CAPACITOR	2,5uF 850V CUBO PMB ICEL	36108
30	1	C20	POLIELSTER CAPACITOR	0,1uF 630V	36149
31	1	C22	CERAMIC CAPACITOR	4,7nF 3KV CERAMIC	86477
32	1	D11	DOUBLE DIODE K.C.	PSMD 150E/12 K.C.	148576
33	1	D14	DOUBLE DIODE A.C.	PSND 150E/12 A.C.	148574
34	1	D16	LED DIODE	VERDE 5MM NORMALE	96004
35	2	D17,D18	DOUBLE DIODE	DSEI2X101-12B IXYS	148101
36	1	EF2	CUSTODIA	4P QUAD/FL HDC04A-ALU	42207
37	1	EL1	SIGNAL LIGHT	M22-LC-W MOELLER	10488
38	1	EM2	CUSTODIA	CUSTODIA 4P QUAD/Q HDC04A-TOLU1PG11G	42206
39	1	ES1	G2R-2 SOCKET	2RF08E OMRON FOR RELE G2R2SN	132098
40	1	ES2	CUSTODIA	PORTALED PANNELLO Ø 5MM BOMBATO INTERNO	96059
41	1	FL1	GRID FILTER	FN 3258-42-47 SCHAFFNER	68111
42	1	F2	FUSE 1AM 6x30	PORTAFUSE ST4HESI 6X32MM PHOENIX	196335
43	2	F3,F4	FUSE 1AT 5x20	PORTAFUSE ST4HESI 5X22MM PHOENIX	196130
44	2	F5,F6	FUSE 2AT 5x20	PORTAFUSE ST4HESI 5X22MM PHOENIX	196130
45	3	F7,F8,F9	FAST FUSE	BUSSMANN 80A	74123
46	3	HR1,HR2,HR3	THERMAL SWITCH	70°C N.C. TYPE TO-220 INTEGRAL	58228
47	1	IA1	MAGNETOTHERMIC SWITCH	FAZ-C50/3	*****
48	1	IMF1	IMPEDANCE	N°4 TOROIDE 43x43x13MM	D10468
49	1	IMF2	IMPEDANCE	IMF-2 MR/B 10/12K-IGBT	176712 EST
50	1	IN1	MAIN SWITCH	P3-63/V/SVB-ZFS68 MOELLER	84163
51	1	MV1	FAN+AIR FILTER	SK3243-100 230VAC RITTAL	*****
52	4	MV2,MV3,MV5,MV6	FAN	120X120MM 220V 15W 4715MS23TB5A	184053
53	1	MV4	FAN	172X150MM 220V 35W 5915PC23TB30	184043

54	1	PD1	THREEPHASE BRIDGE	PSD 162/16	148166
55	3	Q1,Q2,Q3	IGBT MODULE	SKM 200GB 125D SEMIKRON	*****
56	1	Q4	SCR MODULE	SKKH 162-16E SEMIKRON	148316
57	2	RC1,RC2	RC FILTER	0,1uF-100 OHM 630V ARCOTRONIC	68004
58	1	RC2-AUX	RC FILTER	0,1uF 1600V IC + 100ohm 2W	36068+136285
59	1	RL1	RELAY	OMRONG2R-2 2SC 24VDC	132424
60	3	R7,R15,R21	RESISTOR	22 OHM 25W RCH-25	134722
61	1	R19	RESISTOR	500 OHM 25W RCC-1364	134157
62	2	R24,R25	RESISTOR	56R 50W 16X90MM	134093
63	1	R26	RESISTOR	4,7Kohm - 1\4W	136100
64	1	TR1	SERVICE TRASFORMER 250VA	250VA PRI.+20;0;-20:400:460V SEC.0-110V (100VA)-220	170712
65	1	TR2	VACUUM TRANSFORMER		
66	1	U1	SENSOR CURRENT	LEM LA55-SP1	*****
67	1	U6	SECURITY RELAY	440R-N23114 MSR1261T	58849
68	1	XF2	CONNECTOR	HDC-HA-3SS 4P FE	42204
69	1	XF3	CONNECTOR DB25 F	FLAT CONN. 25P FE	42195
70	1	XF4	CONNECTOR	SRCN2A16- 7S FE/PA JAE	*****
71	1	XF5	CONNECTOR	SRCN6A13 3S FE-VO JAE	42139
72	1	XM2	CONNECTOR	HDC-HA-3BS 4P MA	42205
73	1	XM3	CONNECTOR DB25 F	FLAT CONN. 25P MA	42195
74	1	XM4	CONNECTOR	SRCN2A16- 7S MA/VO JAE	*****
75	1	XM5	CONNECTOR	SRCN2A13- 3P MA/PA JAE	42180
76	4	X0.1,X0.2,X0.3,X0.4	SPRING CLIP	ST-10 PHOENIX	196988
77	13	X1.1,X1.2,X1.3,X1.4,X1.5,X1.6,X1.7, X1.8,X1.9,X1.10,X1.11,X1.12,X1.13	SPRING CLIP	ST-2,5 GREY PHOENIX	196748
78	1	X1.14	EARTH SPRING CLIP	ST-2,5/PE PHOENIX	196711
79	1	B5436 D.I	ELECTRONIC CARD	INTERFACE PLC DIGITAL INPUT	200446 EST
80	1	B5436 D.O	ELECTRONIC CARD	INTERFACE PLC DIGITAL OUTPUT	200746 EST
81	1	UP1	CPU 224 SIEMENS	214-1AD23-0XBO CPU DC\DC\DC	127321
82	1	UP2	EM 235 SIEMENS	235-OKD22-0XAO ANALOG MODULE	127236
83	1	UP3	EM 232 SIEMENS	232-OHB22-0XAO ANALOG 2 OUT	127315
84	1	UP4	PROFIBUS DP	277-OAA22-0XAO	127277